THE ASPECT RATIO OF THE TWIN DRAGON IS $1/\varphi$

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Abstract. We show that the geometric aspect ratio of the Twin Dragon equals $1/\varphi$, where $\varphi = (1+\sqrt{5})/2$ is the golden ratio. The result follows by solving the covariance fixed-point equation for the self-similar measure, which coincides with Lebesgue area since the similarity dimension equals 2 and the open set condition holds. The appearance of φ is surprising: the Twin Dragon is defined purely via the Gaussian integer $1+i$, with no pentagonal or Fibonacci structure in its construction. More generally, we derive a closed-form expression for the aspect ratio of the entire one-parameter family $\{f_{1,2}(z) = a(z \pm 1)\}$ for arbitrary $a \in \mathbb{C}$ with $|a| < 1$. The same constant φ also governs the orientation: the major axis of the Twin Dragon makes angle $-\arctan\varphi \approx -58.3°$ with the horizontal.

1. Introduction

The Twin Dragon is a classical plane-filling fractal [6] governed by the Gaussian integer $1+i$: its two defining contractions share the linear part $(1-i)/2$, and their translations are symmetric about the origin. Despite this simple, purely arithmetic construction, the attractor has a non-trivial shape that is not immediately obvious from the definition. Visually the Twin Dragon looks somewhat elongated — neither a square nor a thin sliver — which raises the question of its precise aspect ratio.

A natural quantitative measure of shape for a planar set is the *geometric aspect ratio*: the ratio of standard deviations along the principal axes of its area distribution, or equivalently the square root of the ratio of the eigenvalues of the second-moment (covariance) matrix M of the uniform measure on the attractor. This definition gives the expected answers for elementary shapes: a circle has aspect ratio 1 (all eigenvalues equal); a $p \times q$ rectangle has aspect ratio p/q (eigenvalues $p^2/12$ and $q^2/12$); and a square has aspect ratio 1.

The approach of characterising IFS attractors through moments of their invariant measures was introduced by Vrscay and Roehrig [8] and systematically developed in [9]. The key observation is that the self-similar measure satisfies a linear fixed-point equation in its moments, which can be solved exactly. In favourable cases — when the similarity dimension equals the ambient dimension — this gives the moments of the actual area measure on the attractor.

We apply this method to the Twin Dragon. The covariance matrix turns out to be $M = \frac{1}{5}\begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}$, whose eigenvalues involve $\sqrt{5}$, leading to the aspect ratio

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$1/\varphi$, where $\varphi = (1+\sqrt{5})/2$ is the golden ratio. The appearance of φ in a fractal with no pentagonal geometry is unexpected. Although the method of moments for IFS goes back to [8], the closed-form formula for the aspect ratio of the family $\{a(z \pm 1)\}$ (Theorem 1 below) is, to our knowledge, new.

Discovery. The result was first found empirically using the IFStile software [7], which implements the moment invariants of [8, 9] as part of its algebraic IFS search engine. Among the similarity invariants reported by IFStile for the Twin Dragon, the aspect ratio $1/\varphi$ appeared as a numerical value; the present paper supplies the exact proof and its generalisation to the full one-parameter family $\{a(z \pm 1) : a \in \mathbb{C}, |a| < 1\}$. This is an instance of *experimental mathematics* [2]: a computer experiment suggests a precise conjecture, which is then confirmed by a rigorous proof.

The paper is organised as follows. Section 2 recalls the definition of the Twin Dragon and its IFS. Section 3 derives a closed-form aspect ratio formula for the full one-parameter family $\{f_{1,2}(z) = a(z \pm 1) : a \in \mathbb{C}, |a| < 1\}$. Section 4 applies the formula to the Twin Dragon and proves the main theorem.

2. The IFS

An iterated function system (IFS) [4] is a finite collection of contractions on a complete metric space; by Hutchinson's theorem there exists a unique non-empty compact attractor. For background on self-similar sets and fractal geometry see [5].

The Twin Dragon $A \subset \mathbb{C}$ is the attractor of the IFS [6]

$$A = f_1(A) \cup f_2(A), \qquad f_{1,2}(z) = \frac{1-i}{2}(z \pm 1). \tag{1}$$

Both maps have the same contraction ratio $r = |(1-i)/2| = 1/\sqrt{2}$, so the similarity dimension is $\alpha = -\log 2/\log r = 2$: the attractor has positive area.

3. The general family

Consider the two-map IFS $\{f_{1,2}(z) = a(z \pm 1)\}$ with equal weights $\frac{1}{2}$, for an arbitrary parameter $a \in \mathbb{C}$ with $|a| < 1$. The Twin Dragon is the special case $a = (1-i)/2$, studied in Section 4.

Theorem 1. *Let $a \in \mathbb{C}$ with $|a| < 1$. The invariant (equal-weight) measure μ_a of the IFS $\{f_{1,2}(z) = a(z \pm 1)\}$ has covariance eigenvalues*

$$I_{1,2} = \frac{|a|^2}{2}\left(\frac{1}{1-|a|^2} \mp \frac{1}{|1-a^2|}\right)$$

and aspect ratio

$$\mathrm{AR}^2 = \frac{|1-a^2| - (1-|a|^2)}{|1-a^2| + (1-|a|^2)}. \tag{2}$$

In particular, $\mathrm{AR} = 0$ *if and only if a is real.*

Proof. Let Z be a random variable with distribution μ_a; the fixed-point equation $\mu_a = \frac{1}{2}(f_1)_*\mu_a + \frac{1}{2}(f_2)_*\mu_a$ becomes

$$E[h(Z)] = \tfrac{1}{2}E[h(aZ+a)] + \tfrac{1}{2}E[h(aZ-a)] \tag{3}$$

for every integrable h. Applying (3) to $h(z) = z$ gives $E[Z] = aE[Z]$, hence $E[Z] = 0$ since $|a| < 1$. Applying (3) to $h(z) = z^2$ and expanding:

$$E[Z^2] = a^2 E[Z^2] + a^2, \qquad \text{so} \qquad E[Z^2] = \frac{a^2}{1-a^2}.$$

Applying (3) to $h(z) = |z|^2$:

$$E[|Z|^2] = |a|^2(E[|Z|^2] + 1), \qquad \text{so} \qquad E[|Z|^2] = \frac{|a|^2}{1-|a|^2}.$$

For $Z = X + iY$, the covariance matrix has $\operatorname{tr} M = E[|Z|^2]$ and

$$\det M = \tfrac{1}{4}\big(E[|Z|^2]^2 - |E[Z^2]|^2\big).$$

The eigenvalue discriminant is therefore

$$(\operatorname{tr} M)^2 - 4\det M = \big|E[Z^2]\big|^2 = \frac{|a|^4}{|1-a^2|^2},$$

giving the eigenvalues in the statement. The inequality $|1-a^2| \geq 1-|a|^2$ (triangle inequality) ensures $I_1 \geq 0$, with equality iff a is real. □

Remark 1. *Formula* (2) *shows that* AR *depends on a only through* $(|a|^2, |1-a^2|)$*: the attractor is nearly round when $|1-a^2|$ is large relative to $1-|a|^2$, and degenerates to a segment when $|1-a^2| \approx 1-|a|^2$ (i.e. a nearly real). For every a in the family,* $\mathrm{AR}(a) \leq |a|$*, with equality if and only if a is purely imaginary. Indeed, $|1-a^2| \leq 1+|a|^2$ (triangle inequality), with equality iff $a^2/|a^2| = -1$, i.e. $a \in i\mathbb{R}$. Substituting $|1-a^2| = 1+|a|^2$ into* (2) *gives* $\mathrm{AR}^2 = |a|^2$*. In particular, for the plane-filling sub-family $|a| = 1/\sqrt{2}$ (similarity dimension 2), the maximum* $\mathrm{AR} = 1/\sqrt{2}$ *is achieved by the IFS $\{(i/\sqrt{2})(z \pm 1)\}$, whose attractor is the rectangle $[-1,1] \times [-\sqrt{2}, \sqrt{2}]$ (verified by checking that f_1 and f_2 map it to the top and bottom halves respectively). This rectangle tiles the plane by translation along $[2,0]$ and $[0, 2\sqrt{2}]$. Among named tiles in this sub-family, the Twin Dragon ($a = (1-i)/2$, an algebraic integer satisfying $2a^2 - 2a + 1 = 0$) gives* $\mathrm{AR} = 1/\varphi \approx 0.618$*. Figure 1 shows* AR *over the entire sub-family.*

Remark 2. *A second named tile in the sub-family $|a| = 1/\sqrt{2}$ is the* tame twin-dragon [1, p. 550]*, the attractor of $\{f_{1,2}(z) = a(z \pm 1)\}$ with $a = (1 - i\sqrt{7})/4$. One computes $1 - a^2 = (11 + i\sqrt{7})/8$, hence $|1-a^2| = \sqrt{2}$, and Theorem 1 gives*

$$\mathrm{AR}^2 = \frac{\sqrt{2} - \frac{1}{2}}{\sqrt{2} + \frac{1}{2}} = \frac{2\sqrt{2} - 1}{2\sqrt{2} + 1} = \frac{9 - 4\sqrt{2}}{7} \approx 0.478, \qquad \mathrm{AR} \approx 0.691.$$

This tile arises as the numeration digit set for the complex base $\beta = (1 + i\sqrt{7})/2$ (satisfying $\beta^2 - \beta + 2 = 0$ and $|\beta|^2 = 2$) via $a = 1/\beta$;[1] *it is marked in Figure 1.*

4. The Twin Dragon

For the Twin Dragon the similarity dimension equals the ambient dimension $\alpha = 2$; the open set condition holds, so by [4, §5] the self-similar measure coincides with Lebesgue area on the attractor, and the covariance matrix M describes the actual geometric shape (Figure 2).

[1] The attractor is the set $\{\sum_{k \geq 1} d_k \beta^{-k} : d_k \in \{-1, +1\}\} \subset \mathbb{C}$.

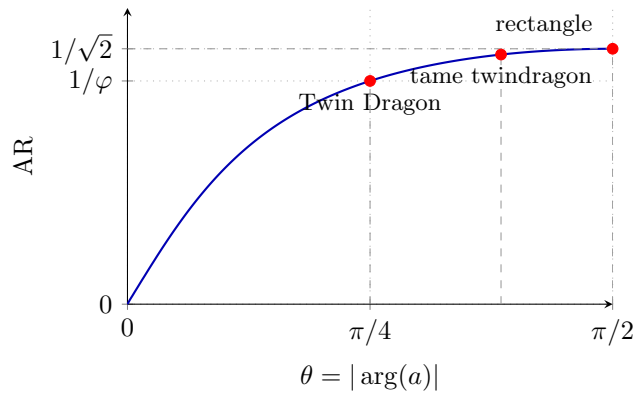

FIGURE 1. Aspect ratio $\mathrm{AR}(\theta)$ of the attractor of $\{a(z \pm 1)\}$ for $|a| = 1/\sqrt{2}$, as a function of $\theta = |\arg(a)|$. The Twin Dragon ($\theta = \pi/4$), the tame twindragon ($\theta = \arctan\sqrt{7}$), and the rectangle attractor ($\theta = \pi/2$) are marked in red.

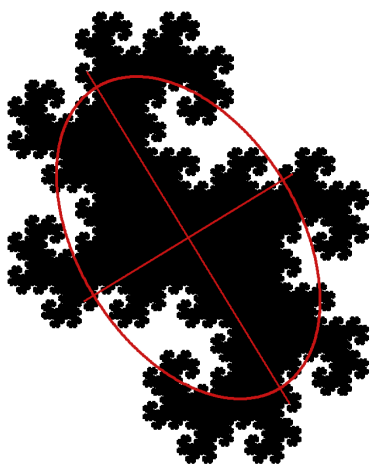

FIGURE 2. The Twin Dragon with its 1.5-sigma principal-axis ellipse (red). The major axis lies along $(1, -\varphi)$ and the minor axis along $(\varphi, 1)$; their ratio of lengths equals the aspect ratio $\mathrm{AR} = 1/\varphi \approx 0.618$.

Substituting $a = (1-i)/2$ into Theorem 1: $|a|^2 = \frac{1}{2}$, $a^2 = -\frac{i}{2}$, $1 - a^2 = \frac{2+i}{2}$, so

$$E[Z^2] = \frac{a^2}{1-a^2} = \frac{-i}{2+i} = \frac{-1-2i}{5}, \qquad E[|Z|^2] = \frac{|a|^2}{1-|a|^2} = 1.$$

The covariance matrix follows via $M_{11} = \frac{1}{2}(E[|Z|^2] + \mathrm{Re}\, E[Z^2])$, $M_{22} = \frac{1}{2}(E[|Z|^2] - \mathrm{Re}\, E[Z^2])$, $M_{12} = \frac{1}{2}\mathrm{Im}\, E[Z^2]$:

$$M = \frac{1}{5}\begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}. \tag{4}$$

The matrix M has $\operatorname{tr} M = 1$ and $\det M = \frac{1}{5}$, so its eigenvalues are

$$I_{1,2} = \frac{1 \mp \frac{1}{\sqrt{5}}}{2}. \tag{5}$$

The geometric aspect ratio — the ratio of standard deviations along the principal axes of M — is

$$\sqrt{\frac{I_1}{I_2}} = \sqrt{\frac{\sqrt{5}-1}{\sqrt{5}+1}} = \sqrt{\frac{3-\sqrt{5}}{2}} = \frac{1}{\varphi}, \tag{6}$$

where $\varphi = (1+\sqrt{5})/2$ is the golden ratio.

Corollary 2. *The attractor of the IFS* $\{f_{1,2}(z) = \frac{1-i}{2}(z \pm 1)\}$ *has covariance matrix* $M = \frac{1}{5}\left(\begin{smallmatrix} 2 & -1 \\ -1 & 3 \end{smallmatrix}\right)$ *and geometric aspect ratio* $1/\varphi = (\sqrt{5}-1)/2 \approx 0.618$, *where* $\varphi = (1+\sqrt{5})/2$ *is the golden ratio.*

Remark 3. *The Twin Dragon is defined by the Gaussian integer* $1+i$, *with no pentagon, decagon, or Fibonacci recurrence in the construction. The emergence of* φ *is therefore unexpected.*

Remark 4. *The golden ratio likewise governs the* orientation *of the Twin Dragon. For* $Z = X + iY$ *with* $E[Z] = 0$, *writing* $E[Z^2] = \rho e^{2i\psi}$ $(\rho > 0)$, *the entries* $M_{11} = \frac{1}{2}(E[|Z|^2] + \operatorname{Re} E[Z^2])$, $M_{22} = \frac{1}{2}(E[|Z|^2] - \operatorname{Re} E[Z^2])$, $M_{12} = \frac{1}{2}\operatorname{Im} E[Z^2]$ *give*

$$M = \frac{E[|Z|^2]}{2} I + \frac{\rho}{2}\begin{pmatrix} \cos 2\psi & \sin 2\psi \\ \sin 2\psi & -\cos 2\psi \end{pmatrix},$$

whose larger eigenvalue has eigenvector $(\cos\psi, \sin\psi)$. *Hence the major axis of any member of the family* $\{a(z \pm 1)\}$ *makes angle*

$$\psi = \tfrac{1}{2}\arg\left(\tfrac{a^2}{1-a^2}\right)$$

with the positive x*-axis. For the Twin Dragon,* $E[Z^2] = (-1-2i)/5$, *so* $2\psi = \arg(-1-2i)$ *lies in the third quadrant with* $\tan(2\psi) = 2$. *Applying the double-angle identity with* $\psi \in (-\frac{\pi}{2}, -\frac{\pi}{4})$, *one gets* $\tan\psi = (-1-\sqrt{5})/2 = -\varphi$. *Thus the major axis has direction vector* $(1, -\varphi)$, *making angle* $-\arctan\varphi \approx -58.3^\circ$ *with the horizontal, and the same constant* φ *governs both the aspect ratio (*$\mathrm{AR} = 1/\varphi$*) and the orientation (*$\tan\psi = -\varphi$*).*

Remark 5. *The number* 5 *— and hence* φ *— can be traced to a single Gaussian prime. By the ergodic theorem for random iteration* [3], μ *is the distribution of the random series*

$$Z = \sum_{k=1}^{\infty} \varepsilon_k \lambda^k, \qquad \lambda = \tfrac{1-i}{2}, \qquad \varepsilon_k \overset{\text{iid}}{\sim} \{\pm 1\},$$

so $E[|Z|^2] = 1$ *and*

$$E[Z^2] = \frac{\lambda^2}{1-\lambda^2} = \frac{-i/2}{(2+i)/2} = \frac{-1-2i}{5}.$$

The ellipticity $\rho = |E[Z^2]| = 1/\sqrt{5}$ *controls the eigenvalue spread:* $I_{1,2} = (1 \pm \rho)/2$. *The value* $5 = \mathbf{N}(2+i)$ *arises as the Gaussian norm of* $2+i$, *which appears because*

$1-\lambda^2 = (2+i)/2$. *Substituting* $\rho = 1/\sqrt{5}$*:*

$$\frac{I_1}{I_2} = \frac{1-\rho}{1+\rho} = \frac{\sqrt{5}-1}{\sqrt{5}+1} = \frac{1}{\varphi^2},$$

where the last step uses $\varphi^2 = \varphi + 1$*. Thus* φ *enters via the Gaussian factorisation* $5 = (2+i)(2-i)$ *in* $\mathbb{Z}[i]$*.*

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